

Minh Vu

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Education

Gettysburg College, Bachelor of Science in Computer Science and Mathematics
Gettysburg, PA

Dec 2026

- **Courses:** Data Structures and Algorithms, Operating Systems, Database System, The Theory of Computation, Computer Vision, Ethical Hacking, Linear Algebra, Real Analysis, Differential Equations, Abstract Algebra
- **Achievements:** 36th place The 2024 ICPC Mid-Atlantic Regional Contest

Experience

Software Engineer Intern, VeneraAI

June 2026 – Present

- Designed a 4-stage clinical transcription pipeline in **FastAPI**: **ffmpeg** normalization, **pyannote** speaker diarization, per-segment offline **Zipformer ASR**, and an LLM post-edit pass that corrects Vietnamese medical spelling
- Built a second cloud-hybrid backend (**ElevenLabs Scribe**) sharing the same DTO contracts as a drop-in alternative

Software Engineer Intern, CMC Global

Nov 2025 – Jan 2026

- Worked with the APAC team to develop a distributed e-commerce platform, building **7 Java Spring Boot** API services supporting payment authorization and courier dispatch
- Engineered **Kafka-based Saga** orchestration with automated rollback and retry mechanisms to maintain data consistency across multi-step payment workflows spanning independent microservices
- Reduced order-status propagation latency from **2–3 minutes** to under **5 seconds** by implementing an **Outbox** event pipeline and real-time **Vue.js** dashboard backed by **PostgreSQL**

Full-stack Developer, Gettysburg College

May 2025 – Jul 2025

- Built and deployed a **React/Express** event-management application on **AWS EC2**, replacing spreadsheet-based tracking with a centralized registration and event workflow backed by **MySQL**
- Reduced event-query response times by **80%** by implementing **Redis** caching for frequently accessed data

Undergraduate Student Researcher, Gettysburg College

Jun 2025 – Jul 2025

- Co-authored a research paper characterizing Li-Yorke chaos and stable periodic orbits through theoretical analysis of discrete dynamical systems
- Developed Python simulations to generate bifurcation diagrams and numerically validate theoretical predictions of chaotic and periodic regimes ([github link](#))

Software Engineer Intern, VinAI

Jun 2024 – Aug 2024

- Worked on the **GuardPro** team, developing an AI-powered object-recognition security camera platform that propagated recognition telemetry across **10+** camera nodes using **REST APIs**, **MongoDB**, **Docker**, and **Kubernetes**
- Engineered **ZooKeeper-based** leader election for distributed AI camera clusters, reducing automated failover latency to under **1.5 seconds** during simulated node outages

Undergraduate Researcher Assistant, Gettysburg College

May 2024 – Jul 2024

- Co-authored “Optimal Play of the All Yellow Zombie Dice Game” published in Springer’s book series ([paper link](#))
- Designed a human-playable strategy achieving 99% alignment with optimal strategy by utilizing a Java simulation

Projects

AlphaGen | Python, Typescript, FastAPI, React, PostgreSQL, Pandas, JWT, Docker Vercel

- Built a full-stack stock analysis platform supporting 4 technical analysis strategies, architecting a **TypeScript/React** frontend and **Python/FastAPI** backend with JWT authentication, deployed on **Fly.io** and **Vercel**
- Engineered a backtesting engine for single and batch historical simulations, integrating **TA-Lib** and **Pandas** with asynchronous processing, **Pydantic** validation, and normalized data models
- Automated market-data ingestion and reporting across intraday, daily, and weekly cadences through a dedicated asynchronous **Python** worker pipeline

Technical Skills

Languages: Go, Java, Python, C++, SQL, TypeScript, JavaScript

Web & Backend: React.js, Node.js, Express.js, FastAPI, Spring Boot, REST APIs, WebSockets

Databases & Data: PostgreSQL, MySQL, MongoDB, Redis

Cloud & Infrastructure: AWS, Docker, Kubernetes, Kafka, RabbitMQ, gRPC, ZooKeeper